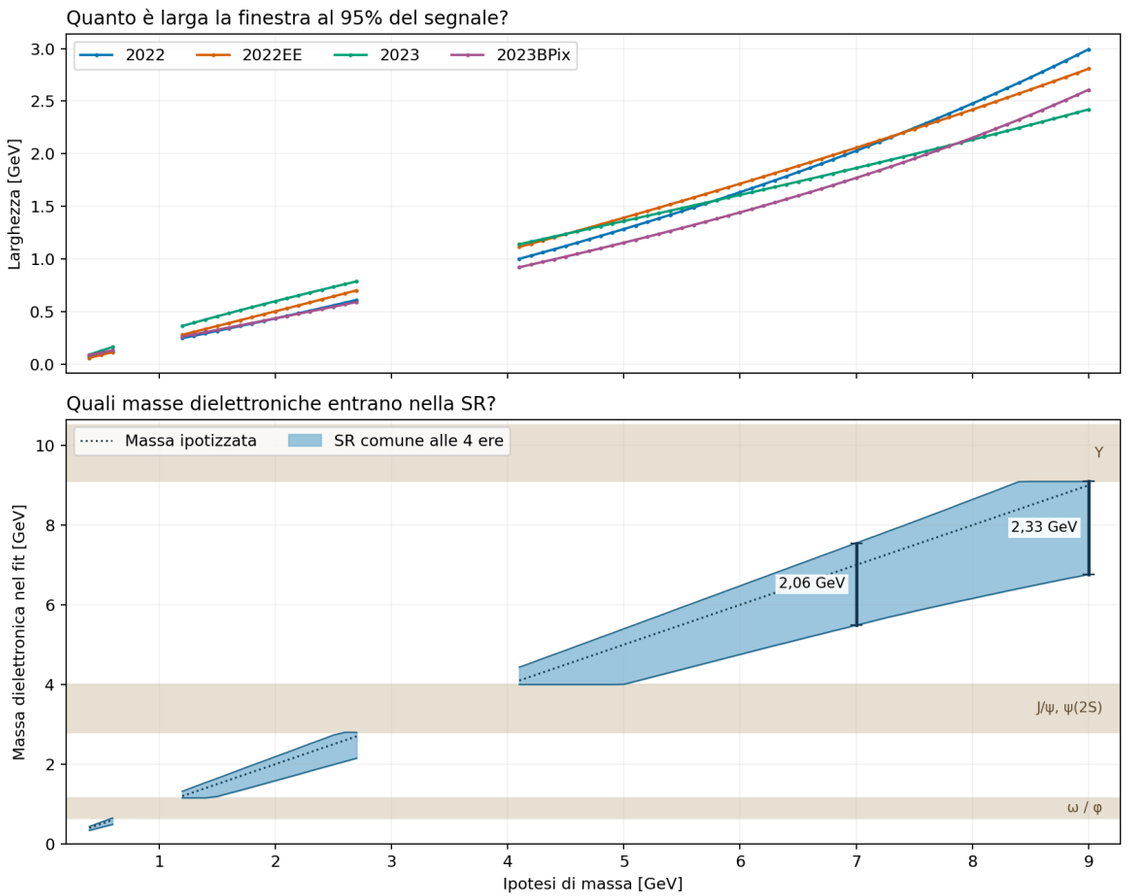
1 Choosing a common signal window

The scan combines 2022, 2022EE, 2023 and 2023BPix. For each mass and era, we find the shortest interval containing 95% of the full parametric double-sided Crystal Ball (DCB) signal. We take the lowest of the four lower edges and the highest of the four upper edges. This common interval covers at least 95% in every era.

The 261106 data use 5 MeV bins. We round the common window outwards to bin edges, then remove bins belonging to any control region (CR). The remaining bins define the signal region (SR). Its signal fraction can therefore fall below 95%. We test 69 masses from 0.4 to 9.0 GeV on a 0.1 GeV grid, skipping hypotheses centred in a CR.



At 0.6 GeV the common interval is 0.49313-0.65505 GeV. Rounding gives 0.490-0.660 GeV, and removing the light-resonance CR leaves 0.490-0.650 GeV (32 bins).

2 Control samples and resonance models

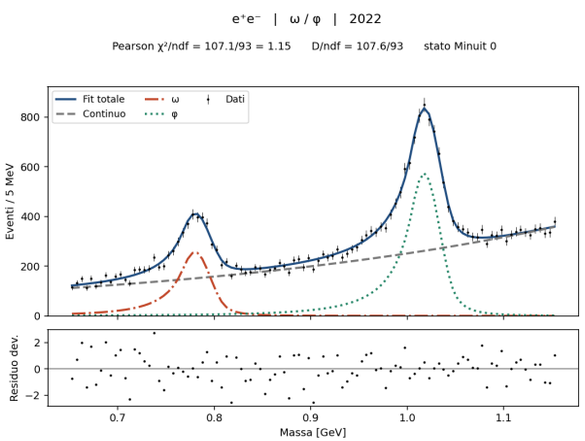
Three local windows group neighbouring SM peaks and include continuum on their flanks. The ranges below are the boundaries used on the 5 MeV grid. Each era is fitted independently in the CR prefit, with one signal-derived DCB per resonance, initialised at its nominal mass. Both peak yields, means and widths float in the prefit.

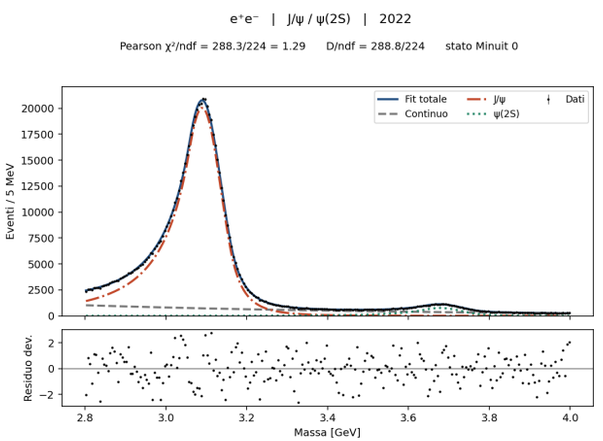
0.650-1.155 GeV | Omega and phi. Exponential continuum. Fix the DCB tails to their signal parametrisation.

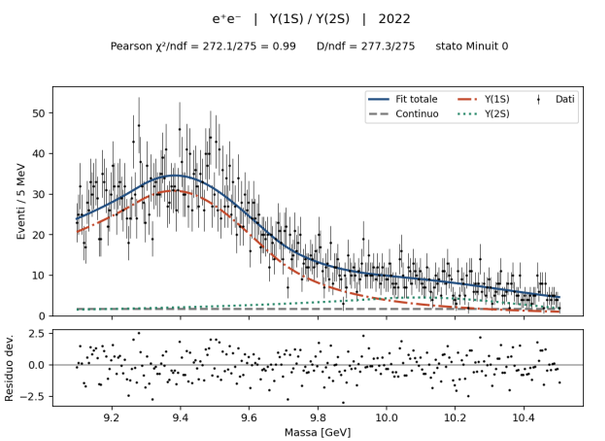
2.800-4.000 GeV | J/psi and psi(2S). A decreasing exponential gives a monotonic, convex continuum. Allow each left and right alpha and n to vary by up to 50% around its signal value in the prefit.

9.095-10.505 GeV | Upsilon(1S) and Upsilon(2S). Constant continuum and fixed signal-derived tails.

The light and Upsilon windows admitted simple continua in the earlier model comparisons. Charmonium retained structured residuals even after freeing its tails. We adopted that shape provisionally. The 2022 prefits below illustrate the fitted components and residuals for all three windows.







3 Simultaneous likelihood for each mass

For each candidate CR, integrate its fitted SM resonance tails over the actual SR bins and sum across its resonances and eras. Rank the CRs by predicted resonance yield. Use the smallest set of one or two CRs covering at least 95% of that yield, as recorded in the mass table. Continuum events do not enter this ranking.

Validation of the CR-selection threshold. The 95% requirement is an operational choice intended to retain nearly all predicted SM resonance tails while limiting the number of control regions. It has no confidence-level interpretation and has not been statistically optimised. The remaining fraction is relative to the resonant contribution alone and does not directly bound the impact on the fitted signal or limits. Validate this choice by adding the next-ranked CR and comparing the fitted signal strength and limits, especially at masses where the selected CR set changes. This stability test remains to be performed.

Each era contributes one SR channel and one channel for each selected CR. Their binned Poisson likelihoods use expected counts integrated over the original bins. At 0.5 GeV the selected light and charmonium CRs give 12 channels in total. At 2.0 GeV charmonium alone gives eight. Each observed bin enters once per era.

A single signal strength r scales the signal in every selected channel and era. The workspace imports the signal shape, yield and ten constrained nuisance parameters from the original signal model: electron ID, reconstruction and trigger, two luminosity terms, electron energy scale and four era-specific resolution terms. Shared nuisance constraints enter once. The original regional signal-rate convention is retained; its absolute normalisation still needs a provenance check.

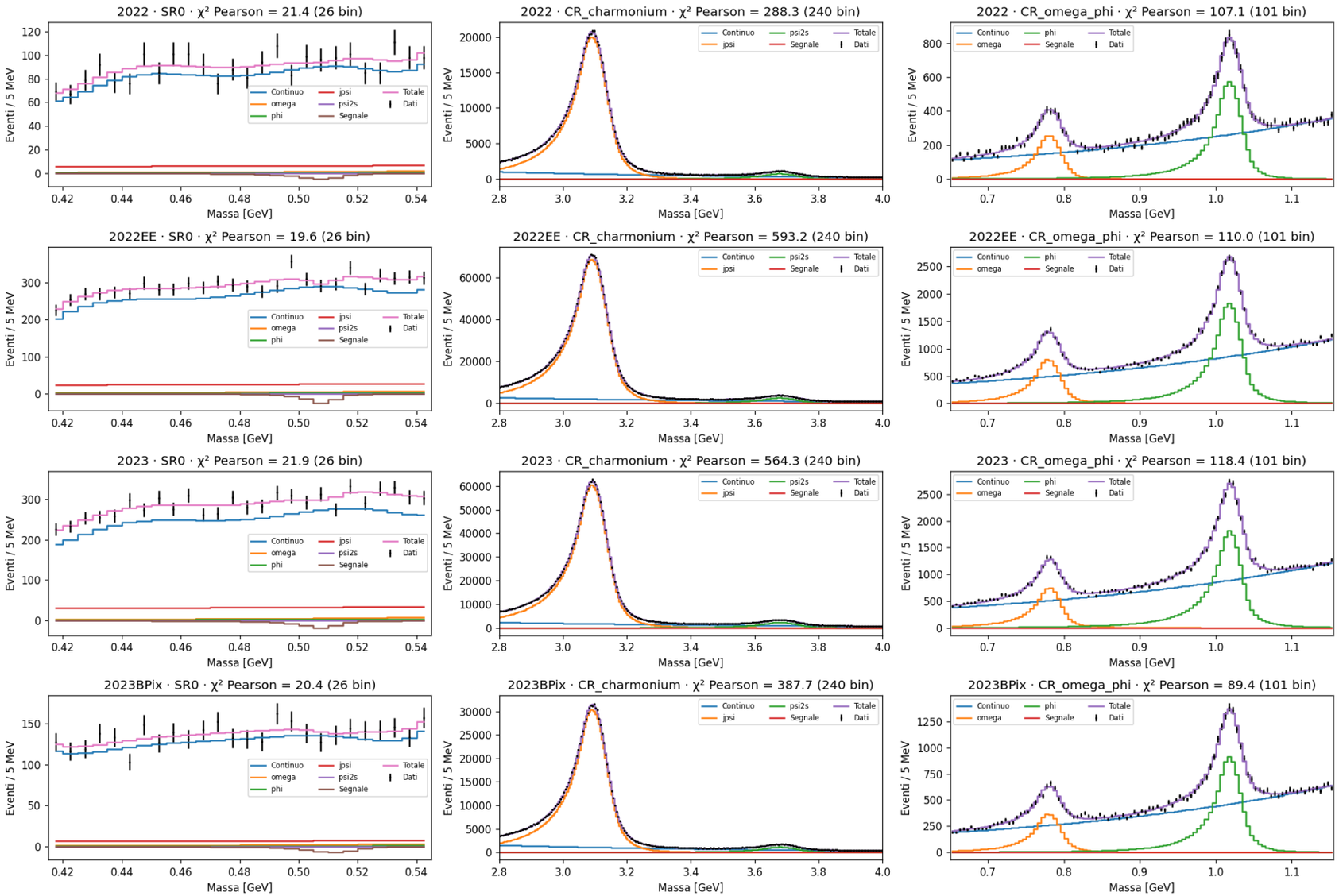
Within each era, each SM peak has a common mean and normalisation across its CR and SR tail. The SM means and CR resonance and continuum yields float. Widths, tails and CR continuum shape parameters are fixed at their adopted prefit values. Their prefit uncertainties are not propagated. The SR continuum shape is profiled.

The baseline limit scan uses a positive degree-six Chebyshev continuum with value matching of the nonresonant density at shared edges. The later envelope tests use value matching of the total physical event density at actual SR/CR contacts. Neither fit matches across gaps. The slope is free. This distinction matters when interpreting the baseline limits alongside the bias tests.

Baseline Combine settings: AsymptoticLimits, Significance and FitDiagnostics use minimizer strategy 0 and MINIMIZER_freezeDisassociatedParams. FitDiagnostics additionally uses robustFit 1, saves shapes and normalisations, keeps failures and allows r from -30 to +30. Toy fits use their own configuration.

4 Low SR: below the phi (0.4-0.6 GeV)

S+B · m = 0.5 GeV · SR Chebyshev 6 · C0 ai bordi comuni · status 0 · covQual 3

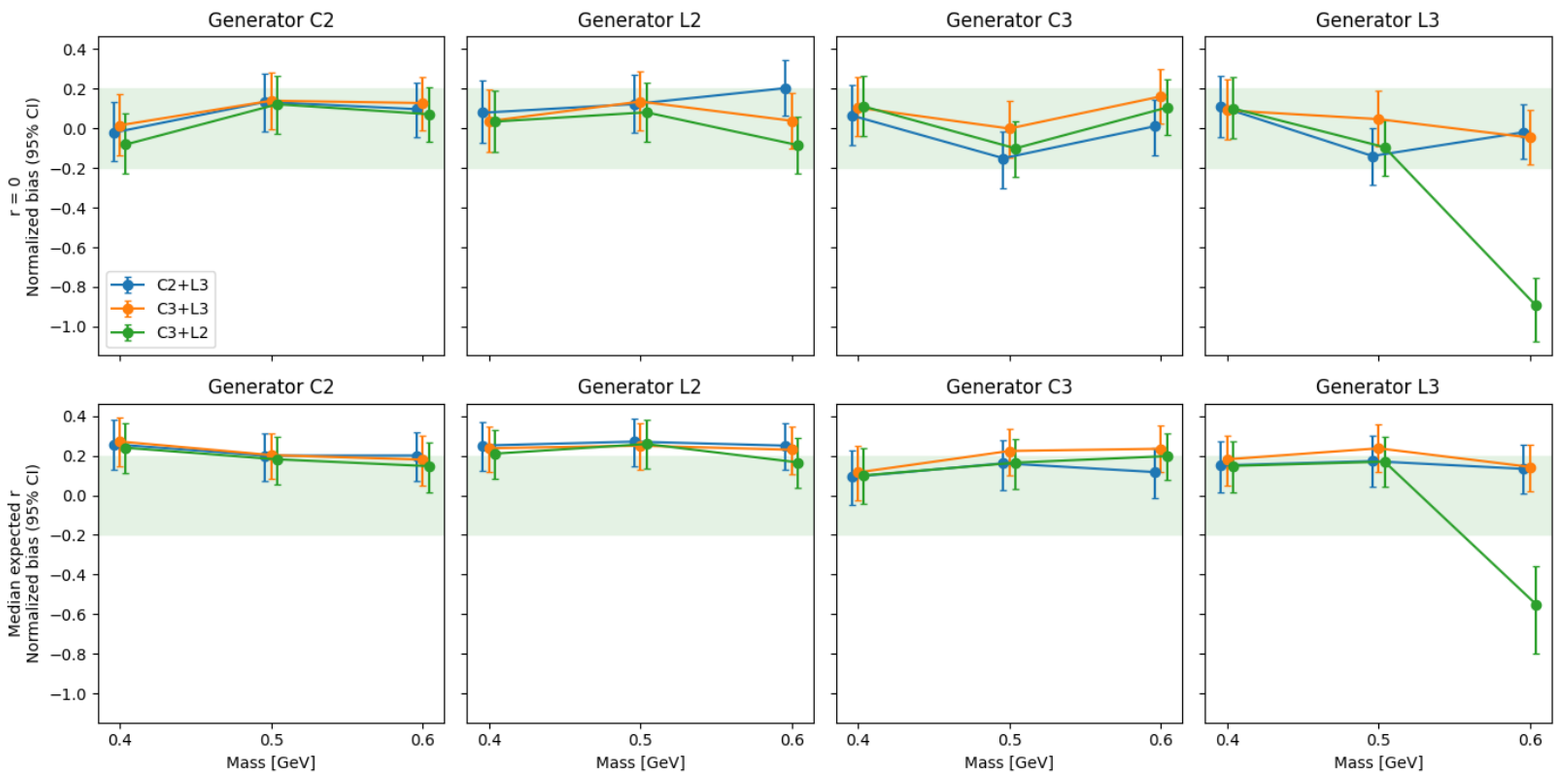


Baseline scan diagnostic from the fit gallery. The four eras use the same effective SR, 0.415-0.545 GeV. The light-resonance and charmonium control samples constrain the SM contributions in the joint likelihood.

4.1 Low-SR envelope selection and bias studies

We generated 200 toys per mass (0.4, 0.5, 0.6 GeV), background generator (Chebyshev or log-polynomial of degree 2 or 3) and injection. The two injections were zero and the mass-specific median expected limit (11.25, 12.234375 and 14.25 respectively). This gives 4,800 distinct toys in 24 configurations.

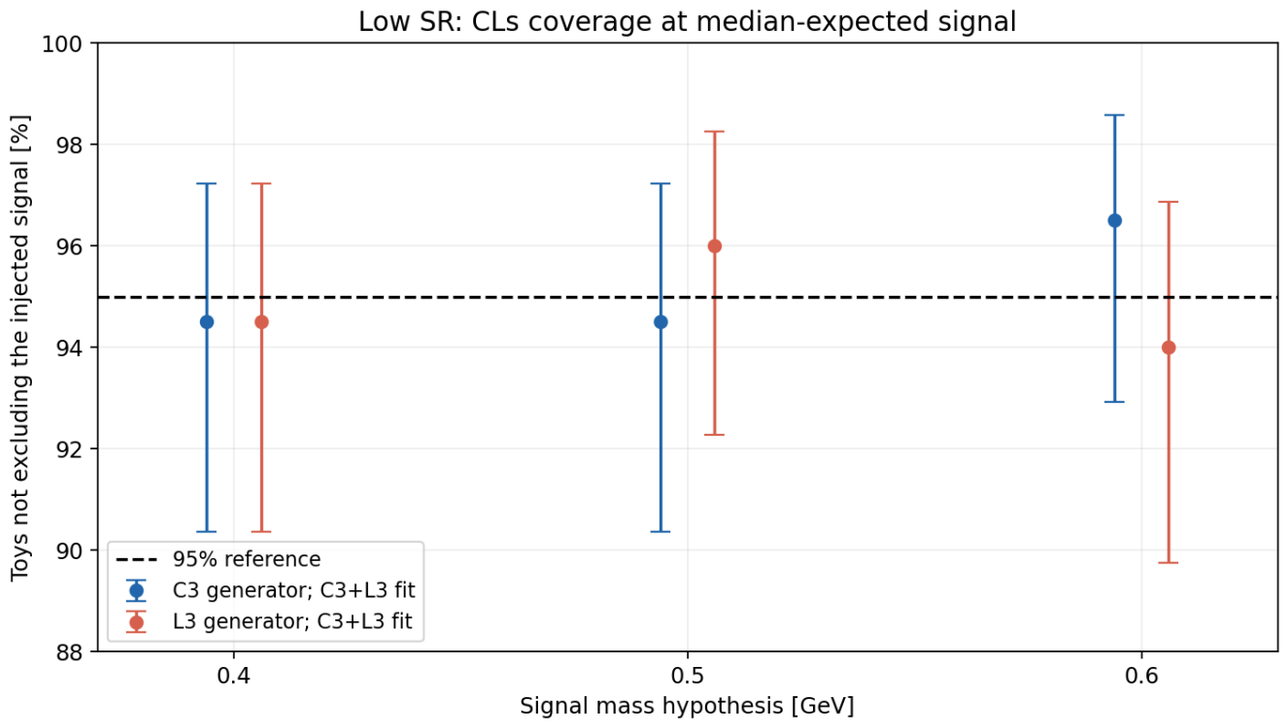
We fit each toy with each candidate branch. For an envelope to enter the comparison both branches must have valid fits. The winner minimises the fitted deviance plus the parameter penalty (+8 for degree 2, +12 for degree 3). The bias is the mean of fitted minus injected r divided by its sample standard deviation. The bars show pointwise 95% bootstrap intervals. The plot compares identical toys valid for all three envelopes (166-200 per configuration).



Recommendation for 0.4-0.6 GeV: C2+L3 or C3+L3. Their maximum absolute biases are 0.271 and 0.271, with 7/24 and 8/24 point estimates outside +/-0.2. Neither meets that threshold everywhere. C3+L2 reaches -0.894 (95% interval -1.073 to -0.753) at 0.6 GeV for L3 generation with zero injection.

4.2 Low SR: coverage with the trigger profiled

C3+L3 envelope; four-era simultaneous SR+selected-CR likelihood. At each mass, inject the median expected signal strength and generate 200 toys with C3 and 200 with L3 (1,200 total). The measured trigger uncertainty remains profiled in the inference.



200 toys per mass and generator; exact 95% binomial intervals. Trigger profiled.

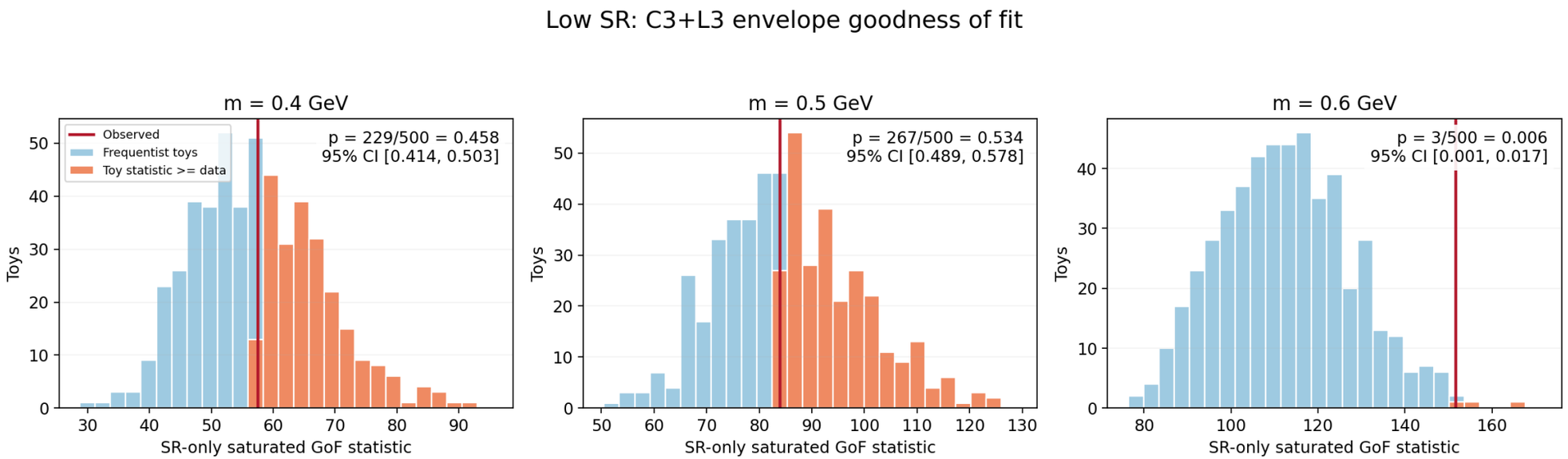
For each toy, evaluate CLs at its injected signal strength. Coverage here is the fraction with CLs >= 0.05: toys that do not exclude the injected signal. Error bars are exact two-sided 95% binomial intervals, not uncertainty bands on the signal limit.

C3 / L3 non-exclusion fractions: 0.4 GeV: 189/200 / 189/200 (94.5% / 94.5%); 0.5 GeV: 189/200 / 192/200 (94.5% / 96.0%); 0.6 GeV: 193/200 / 188/200 (96.5% / 94.0%). All six intervals contain 95%.

No undercoverage is resolved at this toy precision for these generators and injections. This does not establish coverage for arbitrary background shapes, other signal strengths or model misspecification. The coverage result does not override the GoF tension on the following page.

4.3 Low SR: SR-only goodness of fit

Native Combine saturated GoF, with r free in the simultaneous SR+CR fit. All channels enter the fit; the statistic is evaluated in the SR only using the fit/evaluation masks. Use 500 frequentist toys per mass (1,500 total), one Condor job per toy, and the C3+L3 envelope.



Simultaneous SR+CR fit, free r, profiled trigger; statistic evaluated in SR only. 500 toys per mass.

The red line is the observed SR statistic; histograms show the toy distribution. The local p-value is the fraction of toys with a statistic at least as large as observed. Orange bins mark this tail; intervals quantify finite-toy precision.

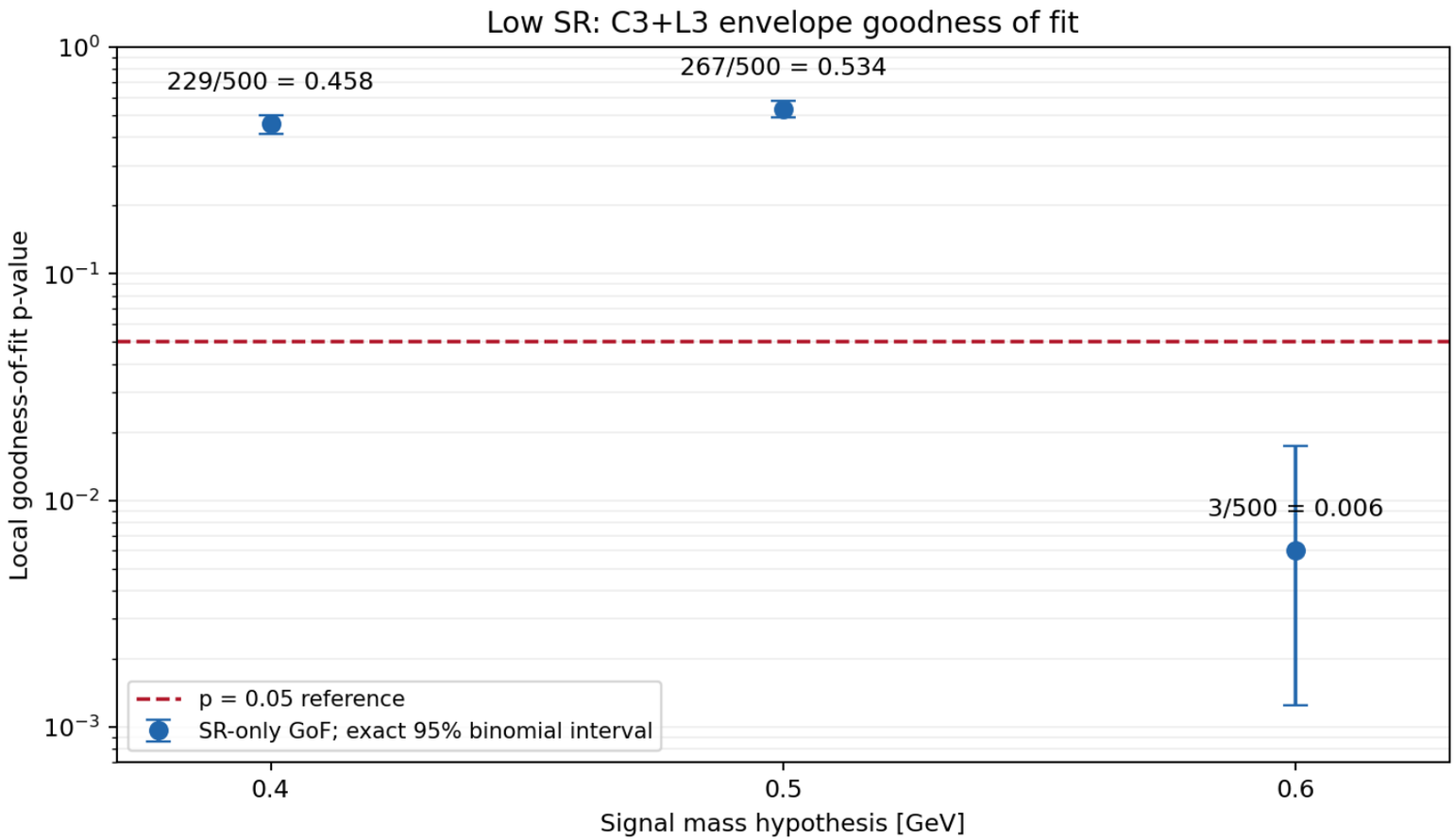
0.4 GeV: observed statistic 57.538; p = 229/500 = 0.458; exact 95% binomial interval [0.414, 0.503].

0.5 GeV: observed statistic 83.956; p = 267/500 = 0.534; exact 95% binomial interval [0.489, 0.578].

0.6 GeV: observed statistic 151.659; p = 3/500 = 0.006; exact 95% binomial interval [0.001, 0.017].

The 0.4 and 0.5 GeV fits show no GoF tension. At 0.6 GeV, p=0.006 flags poor agreement with the tested model: low-SR validation is not uniformly satisfactory. This is a local goodness-of-fit discrepancy, not a signal significance; overlapping windows are correlated. Residual localisation and numerical checks are needed before assigning its cause.

4.4 Low SR: goodness of fit versus mass



Simultaneous SR+CR fit; statistic evaluated in SR only; r free and trigger profiled.
500 frequentist toys per mass. Mass windows are correlated; this is not a signal significance.

Summary of the same 500-toy samples per mass shown on the preceding page. Points are local SR-only GoF p-values, with exact 95% binomial intervals. The dashed line marks p=0.05; it is a reference threshold, not a global significance criterion.

The p-values are 0.458, 0.534 and 0.006 at 0.4, 0.5 and 0.6 GeV, respectively. The 0.6 GeV interval [0.001, 0.017] lies below 0.05. This flags local model disagreement and does not identify its cause or establish a signal excess. The mass windows are correlated.

5 Middle SR: phi to J/psi (1.2-2.7 GeV)

The 16 mass hypotheses span 1.2 to 2.7 GeV in 0.1 GeV steps. At each mass we use the common four-era SR window and the CRs selected in the sliding-window table. The simultaneous fit includes signal in every selected channel and matches the total event density at shared SR/CR edges.

Chebyshev (C) and log-polynomial (L) fits of degrees 2 and 3 at 2.0 GeV provided an initial check of the SR residuals. We then tested whether the envelope could recover a known signal across the mass range. For each degree d, we generated 200 toys per mass with Cd and 200 with Ld. We fitted both branches to each toy and chose the lower penalized deviance among toys with two valid fits. The two branches of each envelope have the same penalty.

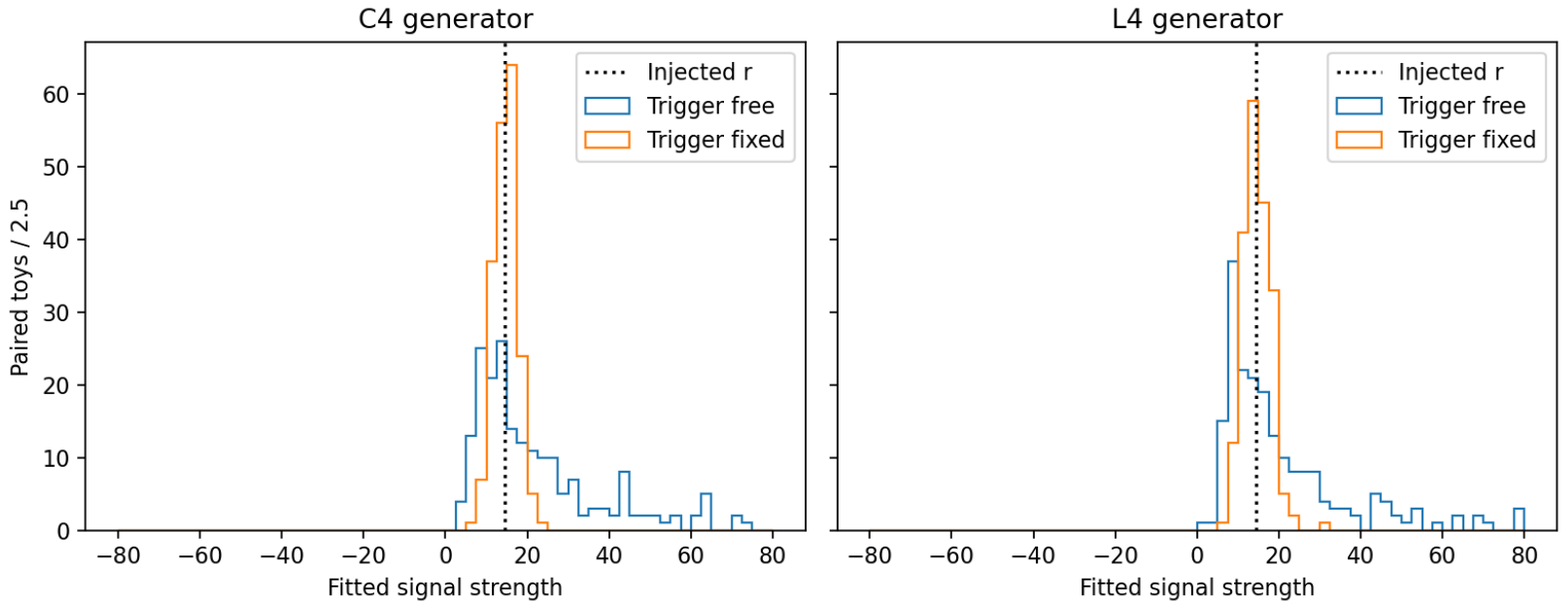
The normalized bias is the mean of fitted minus injected r divided by the sample standard deviation of fitted r. The table counts mass-generator combinations whose point estimate lies outside the target interval [-0.2, +0.2]. The positive injection equals the median expected limit at each mass.

Envelope	Injection	Valid / generated	Outside target	Maximum |bias|
C3 + L3	0	6,352 / 6,400	0 / 32	0.170
C3 + L3	Median expected	6,293 / 6,400	29 / 32	0.490
C4 + L4	0	6,359 / 6,400	0 / 32	0.176
C4 + L4	Median expected	6,277 / 6,400	28 / 32	0.489
C5 + L5	Median expected	5,142 / 5,200	25 / 26	0.494

The cubic and quartic envelopes meet the point-estimate target throughout the background-only tests. With signal injected, both exceed it in most configurations. The quintic test stopped after 13 masses (1.2, 1.8 and 2.7 GeV omitted). At 1.4 GeV its biases are +0.494 and +0.404 for C5 and L5 generation, close to the quartic results in Section 5.1.

5.1 Signal injection at 1.4 GeV and the trigger efficiency

At the median expected signal strength, the quartic envelope overestimates the injected signal on average. We fitted each toy twice with the same simultaneous SR + CR model and injected r = 14.390625. Only CMS_eff_trigger_e changed between fits. It was profiled in the nominal fit and fixed at its injected value of zero in the diagnostic fit. The histograms show the fitted signal strengths for the paired toys.



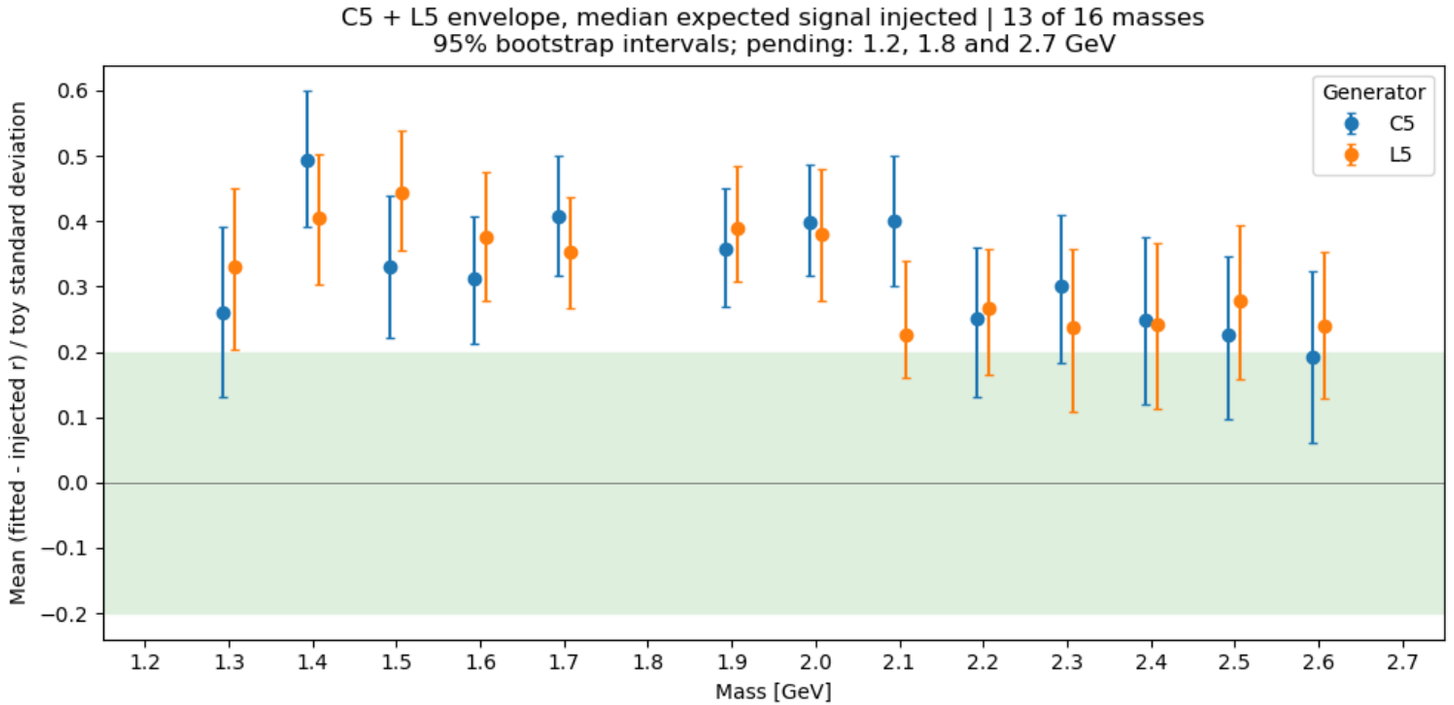
Normalized bias = mean(fitted r - injected r) / sample SD of fitted r. Each row uses the same seeds in both fits.

Generator	Valid pairs	Profiled	Fixed	Change [95% paired bootstrap CI]
Chebyshev 4	195 / 200	+0.489	+0.116	-0.373 [-0.528, -0.222]
Log-polynomial 4	199 / 200	+0.406	+0.071	-0.335 [-0.477, -0.196]

Fixing the trigger reduced the 90th percentile of fitted r from 44.7 to 18.3 (Chebyshev) and from 44.4 to 18.6 (log-polynomial). Six toys failed the two-branch validity requirement in the profiled fit. None of the selected estimates in the paired sample reached a fit boundary. The intervals use 10,000 paired resamples.

The measured trigger uncertainty remains in the nominal likelihood. Coverage requires a separate test of how often the upper limit includes the injected signal. This study measured bias at one mass and injection.

Injected-signal bias and validation of the quartic envelope



Quintic-envelope bias for the 13 completed masses with signal injected. For 14 of the 26 mass-generator combinations the entire pointwise bootstrap 95% interval lies above +0.2.

Coverage has now been tested at all 16 masses (Section 5.3). We generate 200 toys per quartic generator at r = 14.390625 and profile the measured trigger-efficiency nuisance. Combine evaluates asymptotic CLs at the injected r. CLs below 0.05 excludes the true injection. Section 5.3 reports exclusion fractions, binomial intervals and output checks.

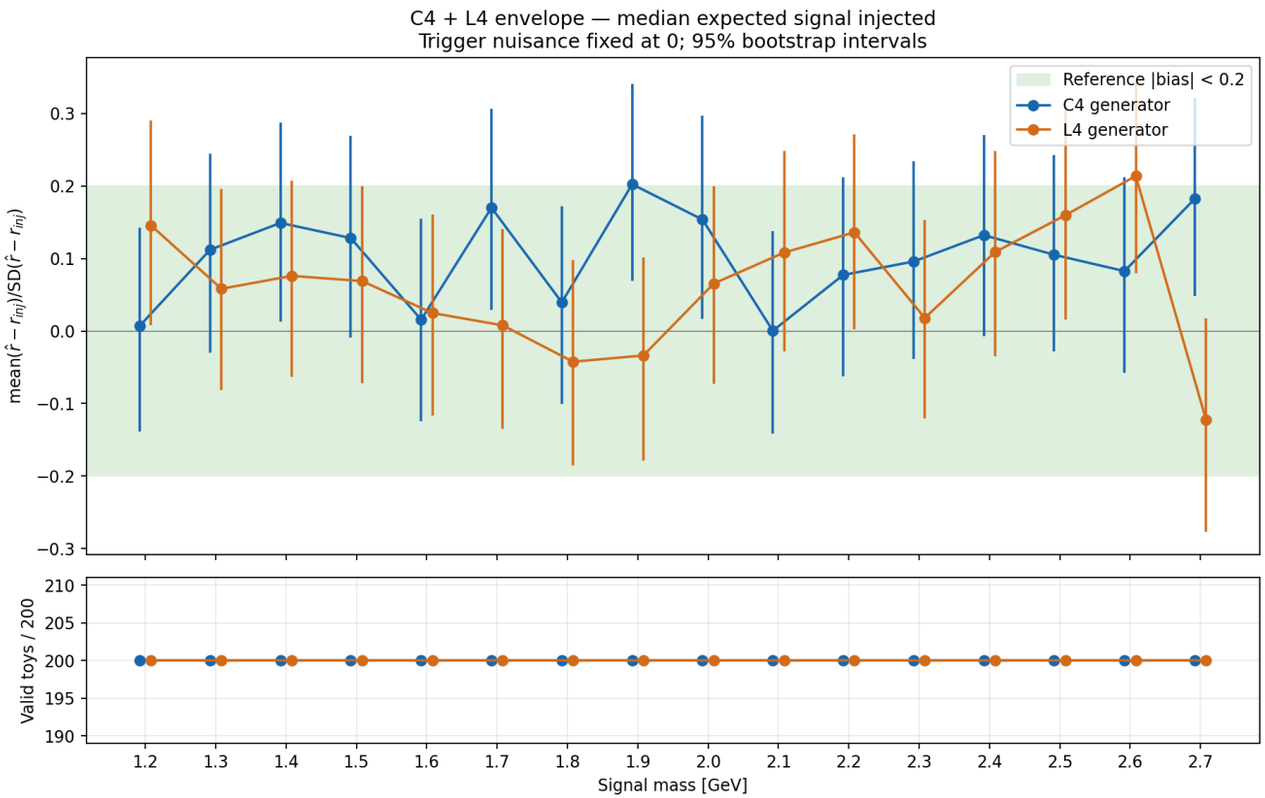
The initial coverage run produced no valid CLs results. The workspace has been corrected to register the shared discrete envelope index in the Combine discreteParams object and exclude it from the continuous nuisance constraint set. Both branch predictions were checked bin by bin. The full rerun is complete.

The full-range fixed-trigger bias results are shown on the next page. The nominal analysis retains the measured trigger uncertainty. Section 5.3 presents the completed coverage test and its implications for envelope selection.

5.2 Fixed-trigger bias across 1.2-2.7 GeV

For each of 16 masses, 200 toys are generated with C4 and 200 with L4 at the mass-specific median expected signal strength. Only CMS_eff_trigger_e is fixed at its injected nominal value, zero, during the refit. The simultaneous SR + selected CR geometry and other fitted nuisance parameters follow the existing quartic study.

All 6400 toys pass the existing validity criteria for both branches. The envelope selects the smaller D+16. Bias is mean(rhat - rinj) divided by the sample standard deviation. Bars show pointwise 95% intervals from 5000 bootstrap resamples.



30/32 mass-generator point estimates fall within +/-0.2. The exceptions are C4 at 1.9 GeV (+0.203) and L4 at 2.6 GeV (+0.214). Several intervals extend beyond the target, so this precision does not establish that every true bias lies within it.

The smaller biases with the trigger fixed support the trigger-profiling explanation. This diagnostic conditions on a known trigger efficiency. Section 5.3 reports the completed coverage test with the measured uncertainty profiled.

5.3 Coverage with the trigger uncertainty profiled

The injected-signal bias study measures the average displacement of the fitted signal strength. Coverage tests a different property of the inference procedure: how often it retains the true injected signal among the values allowed by a nominal 95% CLs exclusion test. A biased point estimate can coexist with adequate coverage.

At each of the 16 masses from 1.2 to 2.7 GeV, we inject a known signal strength equal to the previously calculated median expected limit at that mass. This probes an intensity near the analysis sensitivity. We generate 200 toys with C4 and 200 with L4, for 6400 toys in total. The existing 400 toys at 1.4 GeV are reused.

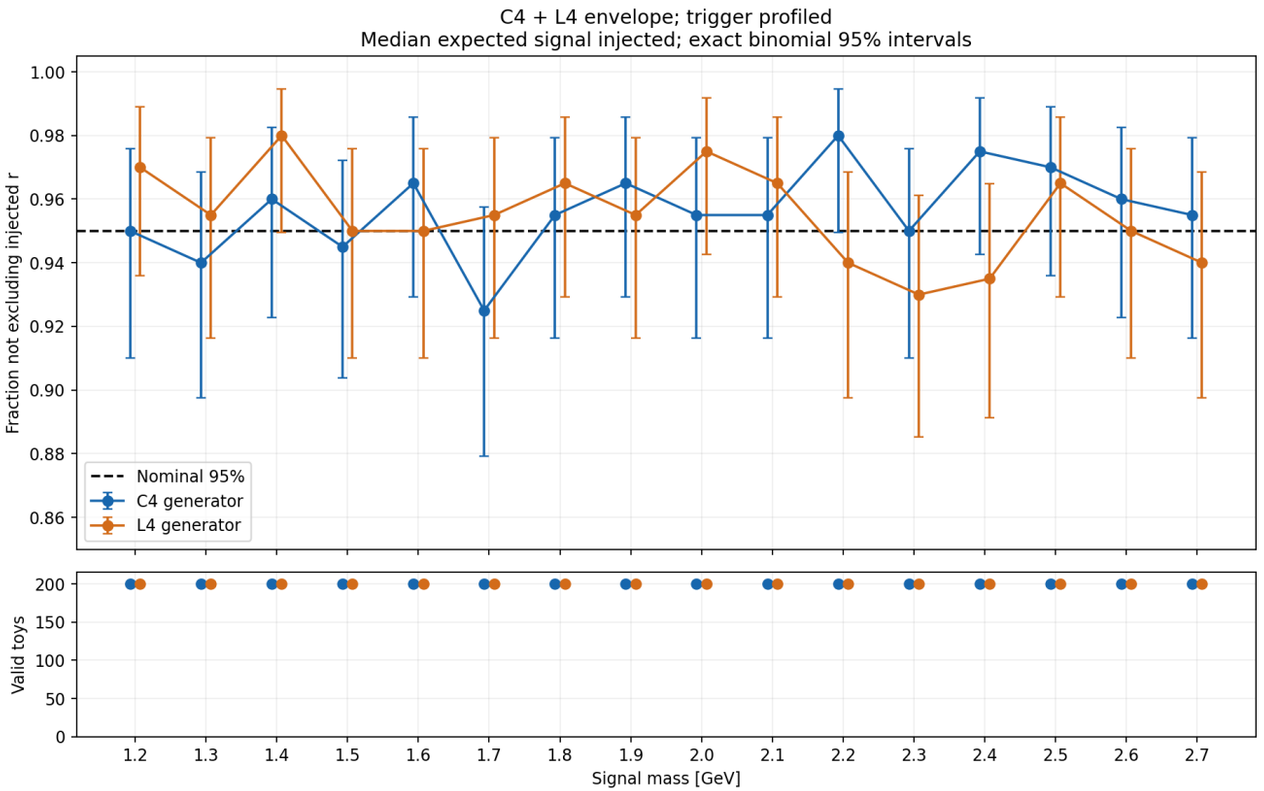
The toys use the saved generator snapshots and the GenerateOnly frequentist-toy configuration. Each toy is analysed with the simultaneous SR plus selected CR model and the shared C4+L4 envelope index profiled across the four eras. The trigger-efficiency nuisance is profiled with its measured constraint. Both branches have the same parameter penalty. Common-boundary C0 relations follow the selected branch, including the CR continuum normalisation at 1.2 and 1.3 GeV.

Combine AsymptoticLimits --singlePoint evaluates CLs at the injected r, using minimizer strategy 0 and MINIMIZER_freezeDisassociatedParams, with rMax = 150. A toy excludes its true injection if CLs < 0.05. The estimated coverage is 1 minus the number of such exclusions divided by 200, calculated separately for every mass and generator.

For example, the C4 sample at 1.4 GeV excludes the true injection in 8 of 200 toys. Its coverage estimate is therefore 192/200 = 96%. The bars on the next page are exact two-sided Clopper-Pearson 95% binomial intervals. They quantify finite-toy uncertainty on the coverage estimate. These intervals replace the approximate Wilson intervals used in the initial chat update.

All 6400 unique seed results are present, with finite CLs in [0,1]. The log scan found no exception, failed-fit or NaN messages. These output and log checks do not prove that every conditional minimisation found its global minimum. The two boundary workspaces passed branch-by-branch expected-count and bin-shape comparisons after including the CR normalisation switch.

Coverage results and implications for the envelope



Coverage estimates range from 92.5% to 98.0%. Every one of the 32 pointwise 95% intervals contains 95%. The lowest estimate is C4 at 1.7 GeV: 15/200 exclusions, or 92.5% coverage, with an exact interval of 87.9-95.7%. Thus no individual configuration shows statistically resolved undercoverage at this precision.

This supports using C4+L4 with the measured trigger uncertainty profiled for the tested region and injections. The positive bias of the fitted-r mean has not translated into resolved undercoverage in this study. An extra systematic uncertainty is not motivated by that bias alone.

The intervals are pointwise, and 200 toys cannot establish exact 95% coverage or rule out modest deficits. The conclusion applies to the C4/L4 generators, saved nuisance truth settings and one injected strength per mass. It does not validate all possible background shapes, signal strengths or simultaneous coverage across the mass scan.

5.4 Goodness of fit in the phi-to-J/psi region

Scope: 16 mass hypotheses, 1.2-2.7 GeV in 0.1 GeV steps, using the adopted C4+L4 envelope and four eras. The signal windows and selected control regions follow the sliding-window table. The fit includes SR and CR data, signal in both, the measured nuisance constraints (including trigger efficiency), and the existing total-density C0 relations at actual common boundaries.

The signal strength r is free in the fit. A shared discrete index chooses C4 or L4, and the continuous nuisance parameters are profiled. This study uses the existing workspace parameter ranges; it does not impose an additional r range on the GoF command.

Why evaluate the SR separately? A global SR+CR goodness-of-fit statistic also tests the CR shapes, including the charmonium description. To examine the SR while retaining the constraints provided by the CR data, all channel masks are set to zero during minimisation, and only the CR masks are set to one when Combine evaluates the statistic. The same mask protocol is used for the observed data and for every toy.

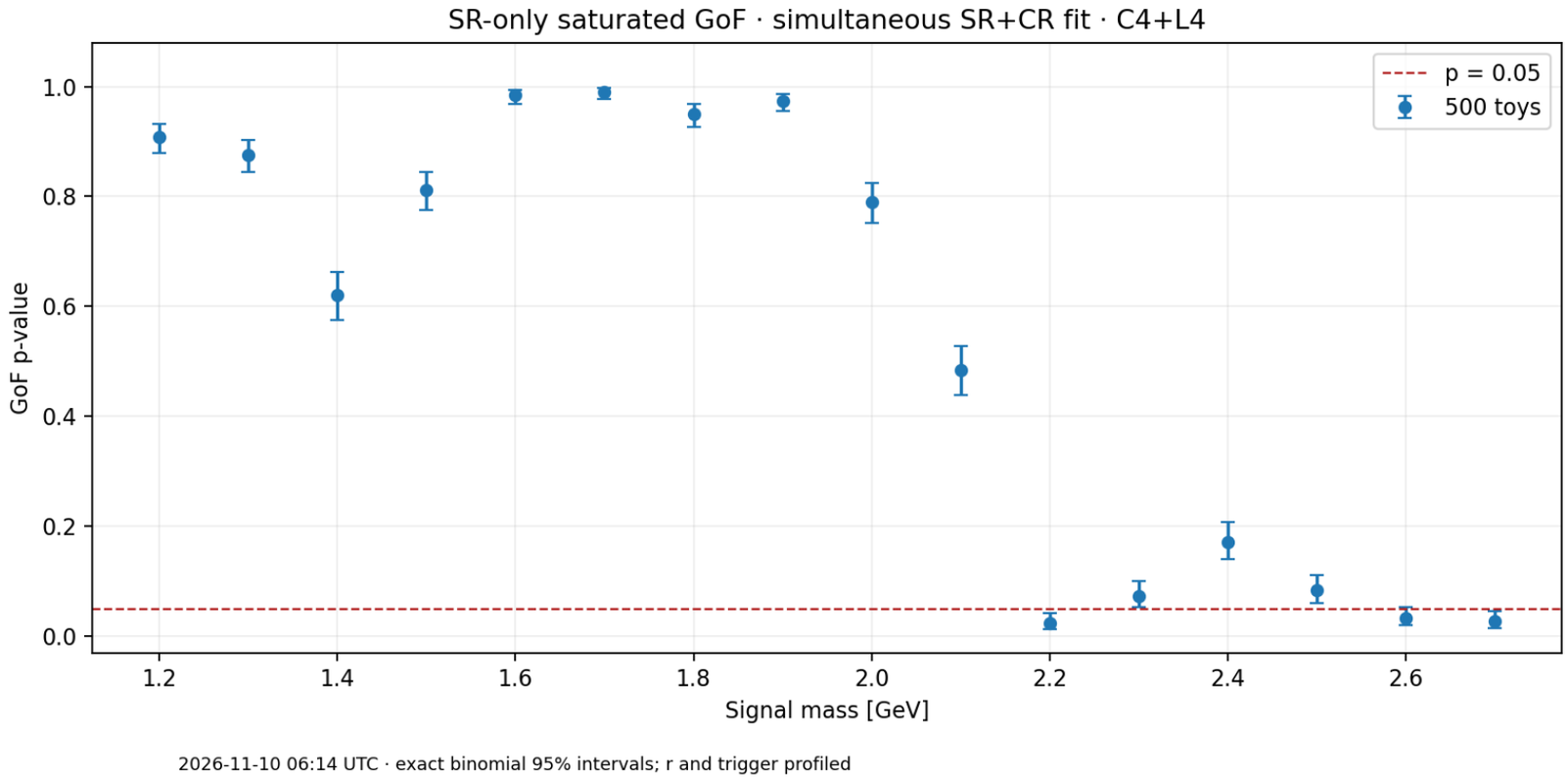
Implementation: native Combine GoodnessOfFit with --algo saturated and --cminDefaultMinimizerStrategy 0, using --setParametersForFit for all masks = 0 and --setParametersForEval for CR masks = 1, SR masks = 0. These settings apply to both nominal and saturated-model stages. This is the native masked saturated-likelihood statistic, not a manually calculated Pearson chi-square; shared auxiliary-constraint handling remains that of Combine.

Calibration: 500 frequentist toys per mass, generated and analysed with -t 1 --toysFreq -s SEED. Each toy runs as its own Condor job and immediately saves a result and exit receipt. The seeds are unique: 91000000 + 1000*i + j, with i = 12,...,27 and j = 0,...,499. All 8000 results are present and finite; the observed statistic is calculated once per mass without the toy-generation options.

The p-value is k/500, where k counts toys with a statistic greater than or equal to the observed value. Error bars are exact two-sided Clopper-Pearson 95% binomial intervals, describing finite-toy precision. The 0.05 reference is a local diagnostic threshold. Overlapping windows make the 16 tests correlated; no global multiple-testing calibration is claimed.

Reproduction: scan_261106/gof_sr_only builds the masked workspaces and evaluates observed statistics; gof_sr_single_toy contains the per-toy runner and full seed manifest. The standalone report script rebuilds the plot and CSV. This GoF tests SR agreement under a simultaneous fit; it does not establish CR adequacy, coverage, or absence of signal-estimator bias. Output completeness alone is not an independent global-minimum audit.

Middle-region GoF results and operational status



Thirteen of the sixteen masses have p >= 0.05. The three values below the local threshold are: 2.2 GeV, 12/500 = 0.024 (95% interval 0.0125-0.0415); 2.6 GeV, 17/500 = 0.034 (0.0199-0.0539); and 2.7 GeV, 14/500 = 0.028 (0.0154-0.0465). The complete numerical table is stored in docs/data/gof_middle_sr.csv.

The calculation is complete, but the outcome is not a uniform GoF pass across the middle region. C4+L4 remains the adopted working configuration; the local tensions at 2.2, 2.6 and 2.7 GeV are retained as qualifications. The mask procedure does not remove the CR data from parameter estimation, and a favourable SR-only p-value must not be interpreted as validation of the CR model.

Operational handover: no further middle-region production is queued by this update. The active model-selection and diagnostic programme moves to the high region, 4.1-9.0 GeV: complete FitDiagnostics, limits and C4/L4 signal-injection studies, and investigate the anomalous observed limits near the Upsilon CR. The recorded middle-region GoF qualifications remain part of the analysis record rather than being reclassified as resolved.

6 High SR: ongoing studies above the J/psi

Scope: 50 hypotheses from 4.1 to 9.0 GeV, in 0.1 GeV steps. C4+L4 is a candidate envelope, not yet a final validated choice. The four-era simultaneous SR+selected-CR likelihood follows Sections 1-3, with signal in all channels, profiled trigger uncertainty and total-density C0 at actual common boundaries.

Completion snapshot: 10 November 2026, 09:07 CET. Counts refer to saved outputs, not a claim that every fit or limit is statistically validated. Figures on the following pages use the same partial-results snapshot.

FitDiagnostics: 48/50 reports. Missing masses: 8.5, 8.6 GeV.

Asymptotic limits: 48/50 high-region points. Missing masses: 7.7, 8.5 GeV.

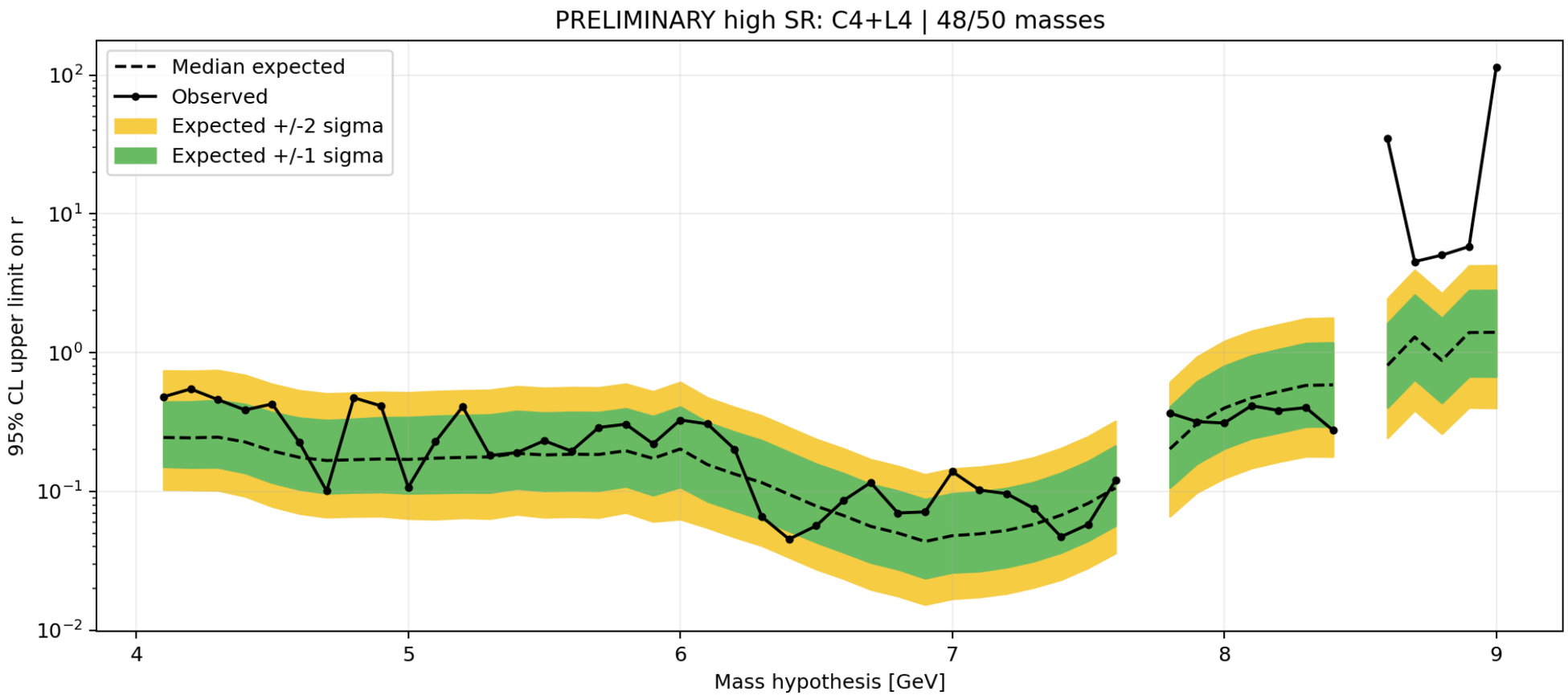
Bias: 24357/40000 toy results received; 24305 valid, 52 invalid, 15643 still missing. Each of 50 masses has two generators (C4, L4), two injections (zero, median expected) and 200 toys per configuration.

Upsilon-boundary diagnosis: 84/108 outputs saved; 24 pending. Conditional likelihood profiles and single-point CLs checks target 8.6 and 9.0 GeV.

Recovery is in progress for the missing fits, limits and generator prefits at 8.5/8.8 GeV. Toy recovery reuses the original seeds and excludes active jobs and saved results. One Condor job per toy is retained. Numerical retries do not relax the model or validity criteria.

No high-region GoF or coverage conclusion is available. The pending programme is to finish and validate the fits and injections, resolve the observed-limit anomalies and then determine whether the candidate envelope is suitable.

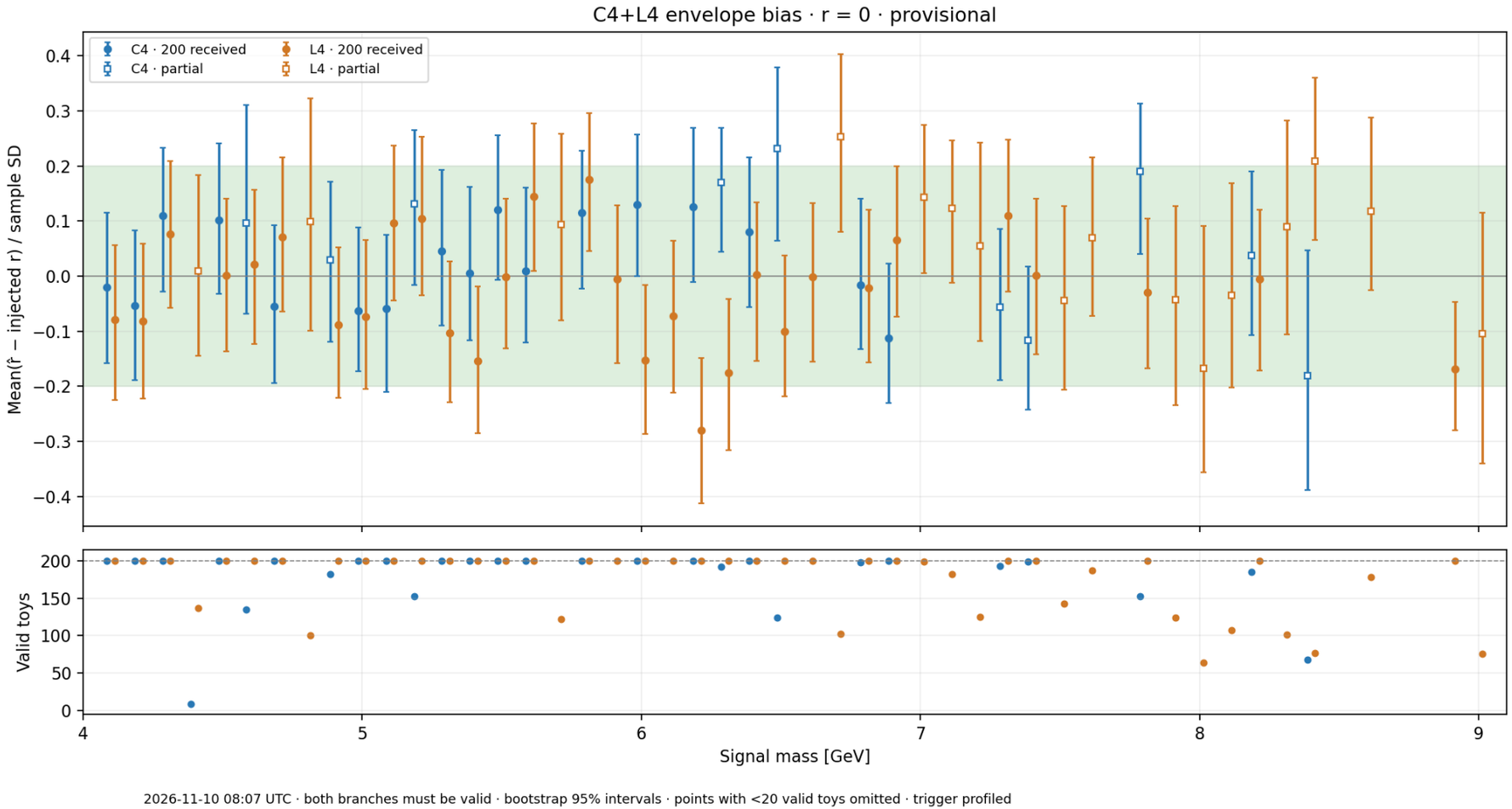
6.1 Preliminary high-region limits



48/50 high-region limits are available; 7.7 and 8.5 GeV are missing and left as gaps. Expected bands are the native AsymptoticLimits quantiles. All included outputs contain the six finite quantiles with ordered expected bands; these checks do not independently validate the profile minima or CLs crossings.

The unusually high observed limits at 8.6 and 9.0 GeV remain under investigation. They are shown without clipping or replacement. These values alone establish neither a physical excess nor a trigger-related explanation. The high-region envelope is still undergoing bias and numerical validation.

6.2 Bias with zero signal

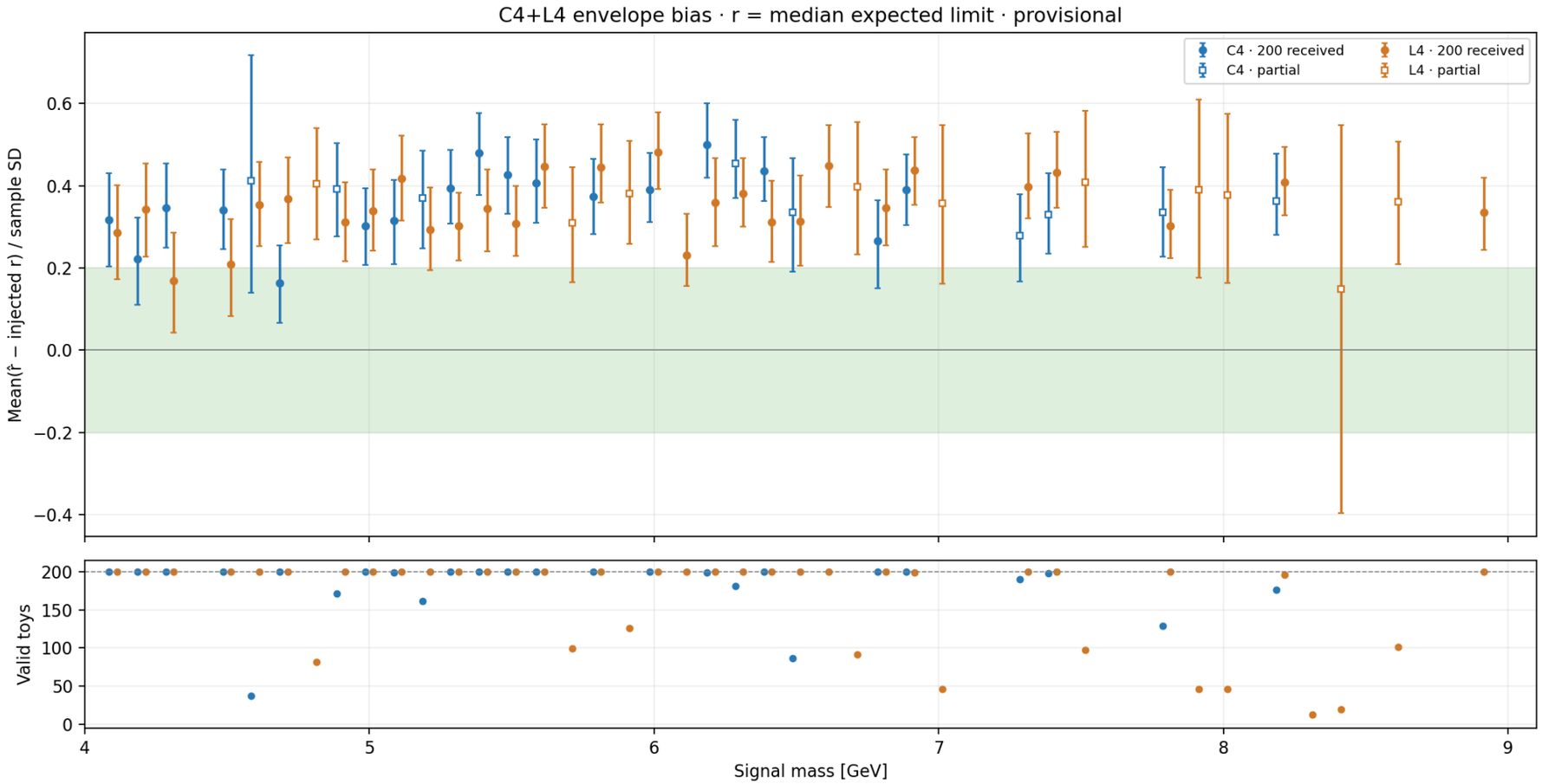


13017/20000 toys received; 13015 valid and 2 invalid; 6983 pending. Complete toy counts at 17/50 masses (both generators), leaving 33 masses incomplete. 47/100 generator-mass samples have all 200 receipts; 73/100 have at least 20 valid toys for plotting. Masses with no receipts: 4 (listed by mass in bias.csv).

Bias = mean(rhat - rinj) / sample SD. Both fitted branches must be valid; the winner minimises D+16. Bars are pointwise 95% bootstrap intervals. Open symbols indicate incomplete samples; full symbols indicate all 200 receipts, not necessarily 200 valid fits. The +/-0.2 lines are diagnostic targets, not confidence limits.

This partial sample does not establish closure. Completion may change the estimates, and invalid fits remain reported separately. The measured trigger nuisance stays profiled, including in the injected-signal study. See bias.csv for per-mass counts and exclusions.

6.3 Bias with median expected signal



11340/20000 toys received; 11290 valid and 50 invalid; 8660 pending. Complete toy counts at 17/50 masses (both generators), leaving 33 masses incomplete. 46/100 generator-mass samples have all 200 receipts; 65/100 have at least 20 valid toys for plotting. Masses with no receipts: 10 (listed by mass in bias.csv).

Bias = mean(rhat - rinj) / sample SD. Both fitted branches must be valid; the winner minimises D+16. Bars are pointwise 95% bootstrap intervals. Open symbols indicate incomplete samples; full symbols indicate all 200 receipts, not necessarily 200 valid fits. The +/-0.2 lines are diagnostic targets, not confidence limits.

This partial sample does not establish closure. Completion may change the estimates, and invalid fits remain reported separately. The measured trigger nuisance stays profiled, including in the injected-signal study. See bias.csv for per-mass counts and exclusions.

7 TODO: planned validation checks

Status: planned, not launched (10 November 2026).
The checks below are deferred work. No new toy campaign is authorised by this documentation update.

1. Validate the C0-only continuum junction at a common SR/CR boundary.
C0 enforces equal physical event density at the boundary; C1 would also enforce equal slope. For a common event selection and no physical threshold at that boundary, a kink is a modelling approximation rather than an expected physical feature. Fixing the CR continuum shape makes C1 a potentially strong transfer of its local slope into the SR.

2. Generate independent smooth-background alternatives.
Use plausible, positive continuum models that are smooth across the actual common boundary, including alternatives outside the fitted C0 family. Preserve the shared SM-resonance and signal construction, the simultaneous SR+CR geometry and all measured nuisance uncertainties. Prioritise mass hypotheses close to the Upsilon CR boundary.

3. Compare C0-only and C1-constrained fits.
Use paired toy samples with zero and median-expected signal injections. Check convergence, signal bias, interval or CLs-limit coverage and sensitivity to the boundary placement. Retain invalid toys and numerical failures in the reporting; use one Condor job per toy with reproducible seeds.

4. State the validation limits and decision before adoption.
Closure when both generation and fitting use the same C0 construction is not sufficient to establish robustness to a smooth true background. Define acceptance criteria before inspecting this comparison. Keep C0 as the current candidate pending validation; do not select the junction prescription to move observed limits into expected bands. The measured trigger uncertainty remains unchanged and profiled in nominal inference.